

Prototype Features - AG Grid Enterprise Support Summary

Overview

This document summarizes the features in the prototype and their support level in AG Grid Enterprise.

Feature Categories & Support Summary

■ Fully Supported by AG Grid (12 features - 31.5%)

These features work out-of-the-box with minimal or no custom code:

- **Hierarchical/Tree Data** - Multi-level data with expand/collapse
- **Cell Editing** - Inline editing of cell values
- **Cell Validation** - Validate edited values
- **Cell Color Coding** - Visual indicators (yellow/orange/red backgrounds)
- **Read-Only Cells** - Lock certain cells from editing
- **Parent-Child Aggregation** - Auto-sum parent rows from children
- **Auto-Update Parents** - Parents update when children change
- **Column Filters** - Filter individual columns with operators
- **Auto-Expand on Search** - Expand rows when search matches
- **Sticky First Column** - Fixed left column while scrolling
- **Column Width Control** - Adjust column widths dynamically
- **Cell Selection & Navigation** - Keyboard navigation, cell selection
- **Virtual Scrolling** - Efficient rendering of large datasets
- **Row Hover Effects** - Visual feedback on row hover

Custom Code Required: 0-20% (mostly styling)

■■ Requires Custom Implementation (26 features - 68.5%)

High Custom Code (80-100% custom) - 11 features

These require significant custom development:

- **Side Panel** (100% custom) - Right panel for cell details/history
- **Undo/Redo** (95% custom) - Undo/redo system with operation tracking
- **Sequential Undo** (95% custom) - Undo notes first, then values
- **Impact Warnings** (95% custom) - Warn when edits affect multiple KPIs
- **Impact Calculation** (90% custom) - Track which cells are impacted when others are edited
- **Edit History** (90% custom) - Track all edits with timestamps
- **Save Functionality** (90% custom) - Save edited values to backend
- **Level Filtering** (85% custom) - Show/hide hierarchy levels
- **Adjustment Notes** (80% custom) - Store and display notes per cell
- **Time Period Filtering** (80% custom) - Filter by Year/Quarter/Month
- **Measure Group Selection** (80% custom) - Filter by KPI sets

Medium Custom Code (60-79% custom) - 9 features

These use AG Grid APIs but need custom logic:

- **Dimension Filtering** (75% custom) - Filter by Account/Category/Product/KPI
- **Time Level Selection** (75% custom) - Switch between Year/Quarter/Month
- **Error Handling** (75% custom) - Validation messages and warnings
- **Global Search** (70% custom) - Multi-term AND search across dimensions
- **Column Width Widget** (70% custom) - Floating widget with slider
- **Loading States** (70% custom) - Loading indicators
- **Percentage Badges** (60% custom) - Show deviation percentages
- **Global Filter Panel** (60% custom) - Right-side filter panel (AG Grid has native Filters Tool Panel, but custom implementation needed for business dimension filtering - Account/Category/Product/Time - and AND logic across filters)
- **Search Highlighting** (60% custom) - Highlight matching cells

Low Custom Code (<60% custom) - 6 features

These mostly use AG Grid with minor customizations:

- **Auto-Expand** (20% custom) - Search-to-expansion logic
- **Cell Color Coding** (10% custom) - Custom color logic
- **Responsive Widths** (10% custom) - Multiplier application
- **Auto-Scroll** (10% custom) - Trigger logic
- **Cell Selection** (5% custom) - Custom CSS styling
- **Chevron Icons** (5% custom) - Custom styling

Key Findings

What AG Grid Provides Natively:

- ■ Tree/hierarchical data structure
- ■ Cell editing and validation
- ■ Column filtering framework
- ■ Aggregation and parent-child calculations
- ■ Virtual scrolling for performance
- ■ Keyboard navigation
- ■ Cell styling and selection

What Requires Custom Development:

- ■■ **Business Logic** (90-100% custom):
 - Impact calculation algorithms
 - Undo/redo system
 - Edit history tracking
 - Adjustment notes storage
- ■■ **UI Components** (60-100% custom):
 - Global filter panel (AG Grid has native Filters Tool Panel, but custom needed for business dimension filtering)
 - Side panel for details
 - Column width widget
 - Loading states
- ■■ **Advanced Features** (60-90% custom):
 - Multi-term global search
 - Dimension filtering
 - Time period filtering
 - Save functionality

Development Effort Estimate

Low Effort (Native AG Grid):

- **12 features** - Use AG Grid APIs directly
- **Estimated Effort:** 1-2 weeks

Medium Effort (Custom on AG Grid):

- **15 features** - Build custom components using AG Grid APIs
- **Estimated Effort:** 8-12 weeks

High Effort (Mostly Custom):

- **11 features** - Significant custom development
- **Estimated Effort:** 12-16 weeks

Total Estimated Development Time: 20-30 weeks (5-7.5 months)

Recommendation

Use AG Grid Enterprise as the foundation because:

- **Core Grid Features** (31.5%) are fully supported - saves significant development time
- **Custom Features** (68.5%) can leverage AG Grid's APIs and extensibility
- **Performance** - AG Grid's virtual scrolling handles large datasets efficiently
- **Maintainability** - Less custom code for core grid functionality

Development Strategy:

- Start with AG Grid's native features (tree data, cell editing, filtering)
- Build custom components for business logic (impact calculation, undo/redo)
- Use AG Grid's APIs as foundation for advanced features

Conclusion

Overall Support: ~95% of features can be implemented using AG Grid Enterprise

- **Native Support:** 12 features (31.5%)
- **Custom Implementation:** 26 features (68.5%)
- High custom (80-100%): 11 features
- Medium custom (60-79%): 9 features
- Low custom (<60%): 6 features

AG Grid Enterprise provides a solid foundation, but the prototype requires significant custom development for business-specific features like impact calculation, undo/redo, and adjustment notes.